AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Model Research and System Architecture

1. Model Research

The proposed system uses Machine Learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI to analyze workforce data and provide useful insights to HR teams.

1.1 Employee Attrition Prediction Model

- Models: Logistic Regression, Random Forest, XGBoost
- Input: Age, experience, salary, job satisfaction, overtime, performance, promotion history, etc.
- Output: Low, Medium, or High attrition risk.
- Purpose: Helps HR identify employees who may require retention strategies.

1.2 Employee Performance Prediction

- Models: Linear Regression, Random Forest, Gradient Boosting
- Input: Performance history, attendance, training, productivity, project involvement, feedback, etc.
- Output: Predicted performance score or performance category.
- Purpose: Supports performance management and workforce planning.

1.3 Employee Skill & Talent Intelligence Model

- Techniques: NLP, text embeddings, similarity analysis
- Input: Resume, employee profile, skills, certifications, training records, job descriptions.
- Output: Skill matches, skill gaps, and recommended learning areas.
- Purpose: Helps organizations plan employee development and internal mobility.

1.4 LLM + RAG Model for Workforce Querying

- LLM: Used for natural-language understanding and response generation.
- RAG: Retrieves relevant information from organizational documents and databases.
- Example query: “Which departments have the highest attrition risk?”
- Output: Data-driven answers, summaries, and recommendations.
- Purpose: Enables conversational workforce intelligence without requiring users to write SQL queries.

1.5 Agentic AI Recommendation Model

- Agentic AI analyzes predictions and generates recommended HR actions.
- Example workflow: High Attrition Risk → Identify factors → Analyze department → Recommend retention action.
- Possible recommendations: Employee training, career development, compensation review, workload adjustment, mentoring programs, and employee engagement initiatives.
- Purpose: Combines insights from multiple analytics modules and provides actionable recommendations to HR leaders.

2. Proposed System Architecture

The system follows a data → processing → AI/ML → intelligence → dashboard architecture.

HR Data Sources

Employee Data; Performance Data; Recruitment Data; Training/Learning Data; Engagement Surveys; Diversity Data; HR Documents / Job Descriptions

Data Ingestion & Storage

SQL Database / Data Warehouse; CSV / Excel / HR Systems; Document Repository

Data Preprocessing & ETL

Cleaning; Missing Value Handling; Feature Engineering; Data Transformation

ML Analytics

Attrition Prediction; Performance Prediction; Skill Analysis; Workforce Pattern Analysis

NLP / LLM

Text Analysis; Summarization; Query Analysis; Natural-Language Understanding

RAG Pipeline

Document Retrieval; Embeddings; Vector Database; Context Retrieval

Agentic AI Layer

Insight Generation; Risk Analysis; Recommendation Engine; Workforce Planning

Workforce Intelligence Layer

Attrition Risk; Performance Insights; Skill Gaps; DEI Metrics; Recruitment Analytics; Engagement Insights

Dashboard & User Interface

Power BI / Tableau / Streamlit; KPI Cards & Charts; Reports; Alerts; Conversational AI

End Users

HR Leaders; HR Managers; Executives

3. Architecture Flow

Workforce Data → Data Preprocessing → ML Models + LLM/RAG → Agentic AI → Predictions & Recommendations → Interactive Dashboard

4. Simple Architecture Explanation

Layer	Purpose

Data Sources	Collect employee, performance, recruitment, training, engagement and organizational data.
Data Storage	Store structured and unstructured workforce information.
Preprocessing	Clean, transform and prepare data for analysis.
ML Layer	Predict attrition, performance and identify workforce patterns.
NLP/LLM Layer	Understand employee documents and natural-language questions.
RAG Layer	Retrieve relevant organizational information before generating answers.
Agentic AI Layer	Combine insights and generate HR recommendations.
Intelligence Layer	Convert model outputs into workforce KPIs and actionable insights.
Dashboard	Display analytics, predictions, trends and recommendations.
Users	HR managers, HR leaders and executives.

5. Conclusion

The proposed AI-Powered Workforce Analytics & Talent Intelligence Dashboard integrates machine learning, NLP, LLMs, RAG, and Agentic AI to transform workforce data into actionable business intelligence. The system can help organizations predict employee attrition, evaluate performance, identify skill gaps, monitor workforce trends, and provide intelligent recommendations for workforce planning and employee engagement.